

Practical 1

Aim: Implement basic operation on arrays: insertion, deletion, updating and searching elements.

Source-code:

```
#include<iostream>
using namespace std;
int main()
{
    int arr[100],n,pos,value;
    cout<<"Enter number of elements:";
    cin>>n;
    cout<<"Enter elements:\n";
    for(int i=0; i<n; i++){
        cin>>arr[i];
    }
```

//INSERTION

```
cout<<"\n--Insertion Operation --";
cout<<"\nEnter position to insert(0 to "<<n<<"):";
cin>>pos;
cout<<"Enter value to insert:";
cin>>value;
for(int i=n; i>pos; i--){
    arr[i]=arr[i-1];
}
arr[pos]=value;
n++;
cout<<"\nArray after insertion:\n";
for(int i=0; i<n; i++){
    cout<<arr[i]<<" ";
}
```

//DELETION

```
cout<<"\n\n--Deletion Operation--";
cout<<"\nEnter position to delete(0 to "<<n-1<<"):";
cin>>pos;
for(int i=pos; i<n-1; i++){
    arr[i]=arr[i+1];
}
n--;
cout<<"\nArray after deletion:\n";
for(int i=0; i<n; i++){
    cout<<arr[i]<<" ";
}
```

//UPDATION

```
cout<<"\n\n--Updation Operation--";
cout<<"\nEnter position to update(0 to "<<n-1<<"):";
cin>>pos;
cout<<"Enter new value:";
cin>>value;
arr[pos]=value;
cout<<"\nArray after updation:\n";
for(int i=0; i<n; i++){
    cout<<arr[i]<<" ";
}
```

//SEARCHING

```
cout<<"\n\n--Searching Operation--";
cout<<"\nEnter value to search:";
cin>>value;
bool found=false;
for(int i=0; i<n; i++){
    if(arr[i]==value){
        cout<<"Value found at position: "<<i<<endl;
        found=true;
        break;
    }
}
if(!found){
    cout<<"Value not found in the array.\n";
}
return 0;
}
```

Output:

Enter number of elements:4

Enter elements:

2 6 9 7

--Insertion Operation --

Enter position to insert(0 to 4):2

Enter value to insert:1

Array after insertion:

2 6 1 9 7

--Deletion Operation--

Enter position to delete(0 to 4):1

Array after deletion:

2 1 9 7

--Updation Operation--

Enter position to update(0 to 3):2

Enter new value:4

Array after updation:

2 1 4 7

--Searching Operation--

Enter value to search:4

Value found at position: 2